Operating Systems Demo Syllabus

A process memory layout typically includes code, data, heap, and stack regio
stack stores function call frames, parameters, and return addresses.

The architecture section explains how stack growth differs from heap growth
process address space.

Stack overflow occurs when nested calls or large local allocations exceed the configured stack region. The syllabus describes the fault as a memory protec failure during execution.

Figure 3 shows the stack growth flowchart, the overflow boundary, and the e raised after the stack limit is crossed.

Example: uncontrolled recursion can trigger stack overflow because each ca another frame onto the stack.

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Context switching saves process state so that the scheduler can resume and process. Scheduling notes relate process control blocks, ready queues, and allocation.

Students are expected to compare recursion depth, stack frames, and excep in system programming questions.